Real-time Seat Occupancy Detection using Computer Vision and Web Dashboard

PROJECT REPORT

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A Project Report submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

In

Electronics and Communication Engineering

By

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25th April 2026

Certificate

This is to certify that the Project entitled "**Real-time Seat Occupancy Detection using Computer Vision and Web Dashboard**" has been carried out by, **HARISANKAR DASH** (2201209641), **ANJALI BHARTI**, (2201209589), **GAURAB PRADHAN**, (2201209638) and **NAYAN RAJ,** (2201209661), **PROVJYUT PRUSTY,** (2221209046) under my guidance and supervision and be accepted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Electronics and Communication Engineering** under **Silicon University**, **Odisha**.

Pradipta Kumar Jena (Sr. Assistant Professor)	**Dr. Amiya Bhusana Sahoo Project Coordinator**	**Dr. Sudhansu Kumar Pati H.O.D, Dept. of EE**

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/sources in our submission. We understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Finally I would like to thank all who's direct and indirect support helped me to complete our project work.

Place: Bhubaneswar

Date: 25th April 2026

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Department Vision, Mission and PEOs

Vision…

To be recognized as a beacon of quality education and research in the field of Electronics and Communication Engineering

Mission…

1. Continually improve the standard of our graduates by having high caliber motivated faculty members together with quality educational programs and facilities in-line with the rapid technological advancements in the field of Electronics and Communication Engineering (Knowledge, Skill and Quality).

2. Provide a balanced regime of quality education that incorporates theoretical and practical education, innovation and creativity as well as freedom of thought and research with emphasis on professionalism and ethical behavior (Professionalism & Ethics).

3. Promote and support research activities over a broad range of academic interests among students and staff for growth of individual knowledge and prepares for continuous learning (Research and Life-long Learning).

Program Educational Objectives (PEOs)

PEO1. Fundamental Knowledge & Core Competence: To utilize the knowledge of mathematics, science and fundamentals of electronics & communication engineering required to become a successful professional and foster complex problem solving ability.

PEO2. Proficiency for the Real World: To acquire the skills to analyze, design, develop, and optimize novel acceptable electronics and communication systems as per the growing needs of the real world.

PEO3. Leadership & Social Responsibility: To exhibit leadership capability with professional, ethical, interpersonal skills, social & economic commitment with a sense of responsibility towards public policies, community services, humanity and environment.

PEO4. Life-long Learning: To grow professionally through continued education & training of technical and management skills, pursue higher studies, and engage in life-long learning.

Abstract

This project presents an IoT-enabled intelligent seat availability detection framework that integrates embedded vision hardware, edge-based image processing, cloud database synchronization, and a custom application interface. The system is designed to monitor and display real-time seat occupancy status in environments such as classrooms and libraries.

The proposed architecture consists of a Raspberry Pi-CAM module for image acquisition, a laptop acting as an edge processing unit, Firebase Realtime Database for cloud synchronization, and a custom application developed using React (web) and Kotlin (Android).

The ESP32-CAM captures real-time images of the seating area and transmits them to the edge device over a local network. Image analysis is performed locally using OpenCV, where computer vision algorithms detect seat occupancy by identifying the presence of individuals within predefined seat regions. Edge image processing ensures reduced latency, minimized bandwidth usage, and improved privacy by transmitting only processed occupancy results rather than raw image data to the cloud.

The processed seat status (occupied or empty) is updated to the Firebase Realtime Database, enabling real-time synchronization across all connected clients. The custom application, developed using React for web-based dashboards and Kotlin for Android platforms, continuously listens for database updates and dynamically reflects seat availability without manual refresh.

This architecture provides a scalable, low-cost, and efficient smart monitoring solution that combines IoT, computer vision, and real-time cloud technologies. The system demonstrates practical implementation of edge computing principles and offers a reliable foundation for smart infrastructure management and resource optimization in modern environments.

Keywords: IoT, Edge Image Processing, Computer Vision, OpenCV, Raspberry Pi-CAM, Real-Time Monitoring, Seat Occupancy Detection, Firebase Realtime Database, Cloud Synchronization, Remote React App Dashboard.

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Chapter 1

Introduction

Introduction

1.1 Introduction

In recent years, the rapid advancement of technology has significantly influenced the development of smart systems for automation and monitoring. One of the key challenges faced in public and private spaces such as libraries, classrooms, auditoriums, and waiting areas is the efficient management of seating arrangements. Traditional methods of monitoring seat occupancy rely heavily on manual observation, which is time-consuming, inefficient, and prone to human errors. These methods fail to provide real-time updates and are not suitable for large-scale environments where quick and accurate information is required.

To overcome these limitations, various automated seat detection systems have been introduced. Most existing systems use hardware-based solutions such as infrared sensors, pressure sensors, and IoT devices installed on individual seats. Although these systems improve automation, they increase installation cost, system complexity, and maintenance efforts. Additionally, such systems are not flexible and become difficult to scale for larger seating arrangements.

With the emergence of computer vision, a new approach has been developed that eliminates the need for physical sensors. Computer vision enables machines to analyze visual data captured through cameras and make intelligent decisions. By using a single camera, multiple seats can be monitored simultaneously, making the system more cost-effective and efficient. This project utilizes computer vision techniques along with a Raspberry Pi to develop a real-time seat occupancy detection system.

The proposed system captures live video through a camera module and processes it using image processing techniques such as grayscale conversion, blurring, and thresholding. The seating area is divided into a grid structure, where each grid represents a seat. The system analyzes pixel intensity within each grid to determine whether a seat is occupied or empty. To improve accuracy, only the central portion of each grid is considered, reducing the impact of noise, shadows, and edges.

The processed data is then displayed on a web-based dashboard that provides real-time updates of seat availability. The interface visually represents seat status using color coding, where green indicates available seats and red indicates occupied seats. The dashboard also displays the total number of seats, occupied seats, and available seats, making it user-friendly and easy to interpret. It can be accessed from multiple devices within the same network, ensuring convenience and accessibility.

This system provides a low-cost, scalable, and non-invasive solution for seat occupancy

detection. It eliminates the need for per-seat hardware installation and offers real-time monitoring capabilities. The implementation of such a system can significantly improve space utilization, enhance user experience, and contribute to the development of smart infrastructure. Below Table 1.1 showing detailed comparison between various popular Biometric methods.

Method	Hardware Required	Cost	Accuracy	Maintenance	Scalability
Manual Monitoring	None	Low	Low	Low	Poor
IR Sensor-Based System	IR Sensors per seat	High	Medium	High	Limited
Pressure Sensor System	Pressure pads	High	High	High	Limited
IoT-Based Smart System	Sensors + Network	Very High	High	Very High	Moderate
Computer Vision System	Camera + Processor	Low	High	Low	Excellent

Table 1.1. Comparison of Seat Occupancy Detection Methods.

1.1.1 Introduction to Seat Occupancy Detection Systems

In modern libraries and study environments, managing seating arrangements efficiently has become a significant challenge. With the increasing number of users, especially during peak hours, it becomes difficult for students to find vacant seats quickly. Traditional methods of monitoring seat availability rely on manual observation, which is time-consuming, inefficient, and prone to human error.

To overcome these limitations, Seat Occupancy Detection Systems have been introduced. These systems use advanced technologies such as sensors, cameras, and image processing techniques to automatically detect whether a seat is occupied or vacant. The primary goal of such systems is to provide real-time information about seat availability, thereby improving user convenience and optimizing resource utilization.

The proposed system uses a vision-based approach, where a camera captures real-time images of the seating area, and a processing unit analyzes these images to determine occupancy status. This eliminates the need for manual supervision and ensures continuous monitoring.

1.1.2 Growth of Smart Monitoring Systems

With the rapid advancement of technologies like Internet of Things (IoT), Artificial Intelligence (AI), and Embedded Systems, smart monitoring solutions have gained significant importance in recent years. These technologies enable automation, real-time data processing, and remote accessibility, making systems more efficient and reliable.

Vision-based monitoring systems, in particular, have seen tremendous growth due to their flexibility and accuracy. Unlike traditional sensor-based systems, camera-based systems can cover larger areas and provide more detailed information. They are widely used in applications such as surveillance, traffic monitoring, smart homes, and smart libraries.

The increasing demand for smart infrastructure has led to the development of intelligent systems that can automatically monitor and manage resources. In this context, seat occupancy detection systems have become an essential part of smart library management, helping users save time and improving overall efficiency.

1.1.3 Introduction to Image Processing and Camera-Based Detection

Image processing is a technique used to analyze and manipulate visual data captured through cameras. It plays a crucial role in detecting objects, identifying patterns, and extracting meaningful information from images.

In the proposed system, a camera module is used to capture real-time images of the seating area. These images are processed using algorithms that detect the presence of a person. Based on this analysis, the system determines whether a seat is occupied or vacant.

The use of image processing offers several advantages:

- Ability to monitor multiple seats simultaneously
- High accuracy in detection
- Reduced dependency on physical sensors
- Flexibility in implementation

Camera-based detection systems are widely used due to their efficiency and ability to operate in real-time environments. When combined with a processing unit like Raspberry Pi, they form a powerful and cost-effective solution for occupancy detection.

1.1.4. Introduction to Raspberry Pi and System Implementation

The Raspberry Pi is a compact and affordable single-board computer widely used in embedded systems and IoT applications. It provides sufficient processing power to handle image processing tasks and control connected components.

In this project, the Raspberry Pi acts as the central controller of the system. It receives input from the camera module, processes the captured images, and determines the occupancy status of seats. It also controls additional components such as LED lights for illumination.

The implementation of the system involves:

- Capturing images using the camera module
- Processing images using software algorithms
- Detecting seat occupancy
- Displaying or transmitting the results

The combination of Raspberry Pi and image processing techniques makes the system efficient, cost-effective, and easy to deploy.

The proposed Library Seat Occupancy System provides a smart and automated solution for real-time seat monitoring, reducing manual effort and enhancing user convenience in modern library environments.

Chapter 2

Literature Survey

Literature Survey

2.1 Introduction

Occupancy detection systems have gained significant importance with the advancement of sensing technologies, Internet of Things (IoT), and artificial intelligence. Over time, various methods and techniques have been developed to detect human presence efficiently, each differing in terms of accuracy, cost, scalability, and computational requirements. These systems play a crucial role in applications such as smart classrooms, transportation, energy management, and security monitoring. The evolution of occupancy detection methods reflects a transition from traditional sensor-based approaches to modern vision-based intelligent systems.

Some of the methods used in the past years are discussed below:

- Early research by Shun Zhang (1999) focused on Poly-Si TFT image sensor technology, which contributed to the development of advanced image sensing techniques. Although not directly applied to occupancy detection, it laid the foundation for modern camera-based monitoring systems.

- Sava P. Bingulac (1994) introduced adaptive control systems that emphasized real-time system adaptability. These systems are useful in dynamic environments where conditions continuously change, but they are not specifically designed for occupancy detection using vision-based approaches.

- With the rise of IoT, M. Shankar, K. Ramesh, and S. Karthik (2018) proposed a seat occupancy detection system using PIR sensors. This method provided accurate detection and became widely used due to its simplicity and reliability. However, it requires individual sensors for each seat, leading to increased wiring complexity and higher installation costs.

- The introduction of deep learning significantly transformed occupancy detection. YOLOv2 (2016) provided a fast and efficient object detection technique suitable for real-time systems, although it had slightly lower accuracy compared to later models.

- YOLOv3 (2018) further improved detection accuracy and robustness, making it highly effective for real-time human detection. However, it requires high computational power and GPU support, which limits its use in low-cost or embedded systems.

- Fast R-CNN (2015) introduced a region-based approach for object detection, achieving high accuracy in identifying objects. Despite its precision, it is slower and computationally expensive compared to single-stage detectors like YOLO, making it less suitable for real-time applications.

Ref. No.	Author(s) & Year	Method / Technique	Key Idea	Advantages	Limitations
1	Shun Zhang, Member IEEE (1999)	Poly-Si TFT Image Sensor Technology	Development of early image sensing techniques	Foundation for modern vision-based systems	Not directly applicable to occupancy detection
2	Sava P. Bingulac (1994)	Adaptive Control Systems	Real-time adaptive system control	Useful for dynamic environments	Not specific to vision-based detection
3	M. Shankar, K. Ramesh, S. Karthik (2018)	IoT + PIR Sensors	Seat occupancy detection using motion sensors	Accurate detection, widely used	Requires wiring per seat, high installation cost
4	Joseph Redmon, Ali Farhadi (2018)	YOLOv3 Object Detection	Real-time object detection using deep learning	High accuracy, robust detection	Requires GPU, high computational cost
5	Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi (2016)	YOLOv2 Object Detection	Faster and improved real-time object detection	High speed, suitable for real-time systems	Slightly lower accuracy than newer models
6	Ross Girshick (2015)	Fast R-CNN	Region-based convolutional object detection	High detection accuracy	Slower, requires higher computation

Table 1: Analysis of Literature Survey

2.2 Motivation and Objectives

The main motivation behind this study is to develop an efficient, cost-effective, and scalable occupancy detection system. Existing methods either rely heavily on hardware (such as PIR sensors) or require high computational resources (such as deep learning models). Therefore, there is a need to design a system that balances accuracy, speed, and cost while ensuring real-time performance. The objective is to explore and implement a solution that overcomes the limitations of the existing technology

2.3 Conclusion

There has been continuous progress in the field of occupancy detection through the development of various techniques ranging from sensor-based systems to advanced deep learning models. While traditional methods offer simplicity and reliability, they lack scalability and flexibility. On the other hand, modern vision-based approaches provide high accuracy and adaptability but come with increased computational costs. Despite these advancements, no single method fully satisfies all requirements such as low cost, high efficiency, scalability, and real-time operation. Therefore, further research is needed to develop optimized solutions that combine the advantages of both sensor-based and vision-based approaches while minimizing their limitations.

Chapter 3

Proposed Technique Name

Proposed Technique

3.1 Introduction

The proposed system implements a real-time seat occupancy detection mechanism using computer vision techniques and a web-based visualization platform. The system is designed to be lightweight, efficient, and suitable for deployment on low-power devices such as Raspberry Pi, while still providing reliable and real-time performance.

The complete technique consists of multiple stages, including image acquisition, preprocessing, region-based analysis, occupancy detection, multi-threaded execution, and web-based visualization. Each stage is explained in detail below.

Components:

- Raspberry Pi
- Switched Mode Power Supply (SMPS)
- Camera Module
- Switch
- Flash Light
- Jumper wires

The above-mentioned components are used for constructing the proposed model of the Library Seat Occupancy System, where the main components include a Raspberry Pi, Switched Mode Power Supply (SMPS), camera module, switch, flashlight, and jumper wires.

3.1.1 Raspberry Pi

The Raspberry Pi is a compact, low-cost, and highly capable single-board computer developed by the Raspberry Pi Foundation. It is designed to promote computer science education and provide an affordable platform for building embedded systems, IoT applications, and real-time processing solutions.

Despite its small size, the Raspberry Pi functions like a complete computer, capable of running an operating system, executing programs, and interfacing with various hardware components. Due to its low power consumption, portability, and versatility, it is widely used in applications such as automation systems, robotics, image processing, and smart monitoring systems.

In this project, the Raspberry Pi serves as the main processing unit, handling camera input, image processing using OpenCV, and communication with the web dashboard.

The Raspberry Pi is built around a System on Chip (SoC) architecture, which integrates multiple components such as the processor, GPU, memory controller, and input/output interfaces onto a single chip.

3.1.1.1 Main Components of Raspberry Pi

Processor (CPU): Executes all computational tasks and program instructions

GPU (Graphics Processing Unit): Handles image rendering and video processing

RAM: Temporary memory used for running applications

Storage (MicroSD Card): Stores operating system and files

GPIO Pins: Used for interfacing with external hardware

USB Ports: Connect peripherals like keyboard, mouse, and camera

HDMI Port: Used for display output

Networking: Supports Wi-Fi and Ethernet for communication.



Figure 1: Raspberry Pi

3.1.2. Switched Mode Power Supply (SMPS)

3.1.2.1. Introduction

A Switched Mode Power Supply (SMPS) is an electronic power supply unit that efficiently converts electrical power from one form to another using high-frequency switching techniques. Unlike traditional linear power supplies, which dissipate excess energy as heat, SMPS uses switching regulators to provide high efficiency, compact size, and reliable performance.

SMPS is widely used in modern electronic systems such as computers, embedded devices, industrial equipment, and microcontroller-based projects. In this project, the SMPS is used to provide a stable and regulated DC power supply to components like the Raspberry Pi, camera module, and other peripherals.

3.1.2.2. Working Principle of SMPS

The working of an SMPS is based on rapid switching of power devices (transistors) at high frequencies. The input AC voltage is first converted into DC, then switched at high frequency, and finally regulated to the desired output voltage.

The overall working can be divided into the following stages:

1. **Rectification and Filtering**

- The input AC supply (typically 230V AC) is first passed through a rectifier circuit, usually a bridge rectifier.
- The rectifier converts AC into pulsating DC.
- A filter capacitor smooths this pulsating DC to produce an unregulated DC voltage.

2. **High-Frequency Switching**

- The DC voltage is then fed into a switching transistor (MOSFET).
- This transistor rapidly turns ON and OFF at high frequency (typically in kHz range).
- This switching action converts the DC into high-frequency pulses.

3. **Transformer Stage**

- The high-frequency pulses are passed through a transformer.

- The transformer steps up or steps down the voltage as required.
- High-frequency operation allows the transformer to be much smaller compared to traditional transformers.

4. **Output Rectification and Filtering**

- The transformed AC signal is again converted into DC using a rectifier.
- A filter (capacitor/inductor) smooths the output to provide a stable DC voltage.

5. **Feedback and Regulation**

- A feedback circuit continuously monitors the output voltage.
- If the output voltage changes, the switching duty cycle is adjusted.
- This ensures a constant and regulated output voltage regardless of load variations.



Figure 2. Switched Mode Power Supply (SMPS)

3.1.3. Camera Module

A camera module is an image acquisition device that captures still images or continuous video streams. In embedded systems like a Raspberry Pi-based occupancy detection system, the camera acts as the **primary sensing unit**. It captures real-time visual data of the seating area, which is then processed using image processing or computer vision techniques to determine whether a seat is occupied or vacant.

The commonly used camera module (such as Raspberry Pi Camera Module v2) uses a **CMOS (Complementary Metal-Oxide Semiconductor) sensor**, which converts light into electrical signals. These signals are then digitized and transmitted to the Raspberry Pi through a **CSI (Camera Serial Interface)** using a ribbon cable. This interface allows high-speed data transfer, which is essential for real-time video processing.

In the context of a library seat occupancy system, the camera continuously monitors seating zones. The captured images are analyzed using algorithms such as:

- Motion detection
- Background subtraction
- Human/object detection (using OpenCV or AI models)

Based on this analysis, the system determines occupancy status and updates the output accordingly.

Applications in System:

- Detects presence/absence of users
- Provides real-time monitoring
- Enables automation without physical sensors

Advantages:

- Non-contact detection
- Covers multiple seats at once
- High accuracy with proper algorithms

Limitations:

- Performance depends on lighting conditions
- Requires computational processing

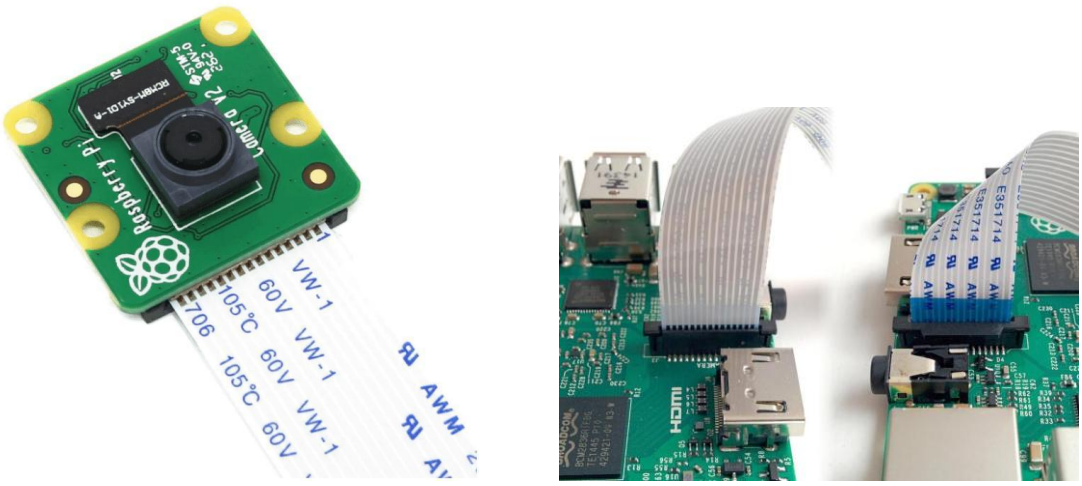


Figure 3: Raspberry Pi Camera Module used for capturing real-time images of the seating area.

3.1.4. Switch

A switch is a fundamental electrical component used to **control the flow of current in a circuit** by either completing (closing) or interrupting (opening) the electrical path. In electronic systems, switches are commonly used for user input and control operations.

In this project, a push-button or toggle switch can be interfaced with the GPIO (General Purpose Input/Output) pins of the Raspberry Pi. When the switch is pressed, it changes the voltage level at the GPIO pin (from LOW to HIGH or vice versa), which is detected by the program running on the Raspberry Pi.

Switches operate on the principle of **mechanical contact**. When pressed, conductive terminals inside the switch touch each other, allowing current to flow. When released, the connection is broken.

Role in the System:

- Power ON/OFF control
- Manual reset of the system
- Triggering image capture or calibration

Types of Switches Used:

- Push button (momentary)
- Toggle switch (latching)

Advantages:

- Simple and reliable
- Low cost
- Easy to interface

Limitations:

- Mechanical wear over time
- Requires debounce handling in software



Figure 4: Electrical Switch

3.1.5 Jumper Wires

Jumper wires are electrical conductors used to establish temporary connections between components in a circuit without the need for soldering. They are widely used in prototyping and development environments.

Jumper wires typically consist of a flexible insulated wire with connectors at both ends. These connectors can be:

- Male-to-Male
- Male-to-Female
- Female-to-Female

In a Raspberry Pi-based system, jumper wires are used to connect components like switches, LEDs, and other peripherals to the **GPIO pins**. They ensure proper signal transmission and power distribution among components.

The internal conductor (usually copper) provides low resistance, allowing efficient current flow. The outer insulation prevents short circuits and protects users from electric shocks.

Role in the System:

- Connects hardware components
- Enables rapid prototyping
- Facilitates easy modifications

Advantages:

- No soldering required
- Reusable and flexible
- Saves time during development

Limitations:

- Loose connections may occur
- Not suitable for permanent installations

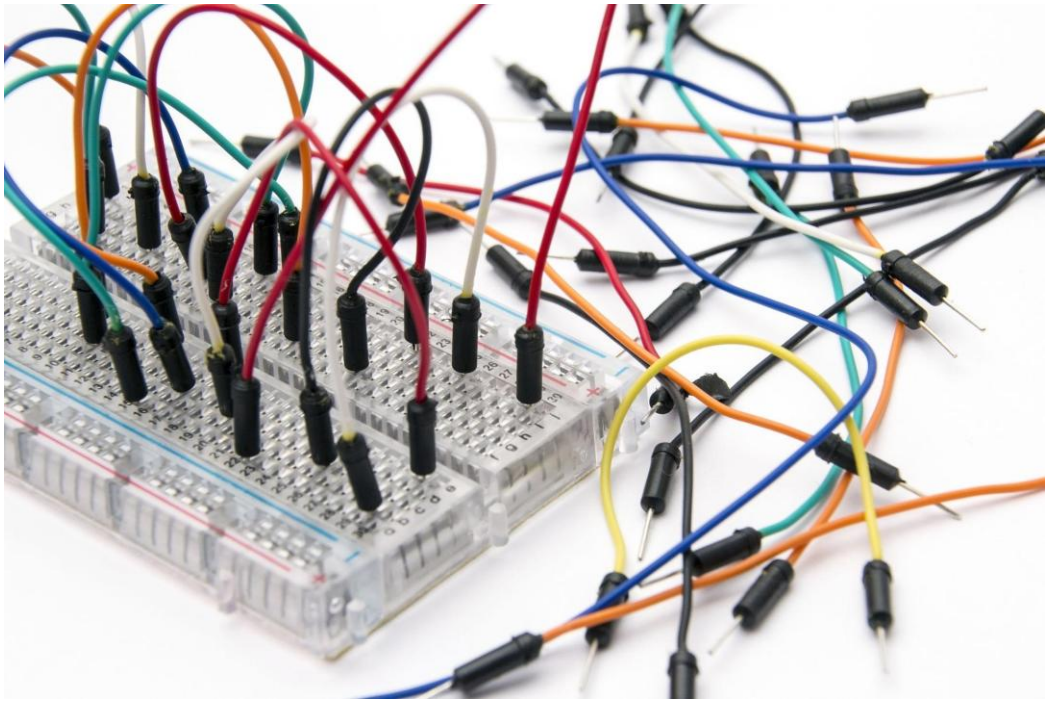


Figure 5: Jumper Wires used for circuit connections.

3.1.6. DC Converter (DC–DC Converter)

A DC–DC converter is a power electronic circuit that converts electrical energy from one DC voltage level to another. It is widely used in embedded systems to provide regulated and stable power to different components.

In practical systems, power sources such as batteries or adapters may not provide the exact voltage required by devices. For example, a Raspberry Pi typically requires a stable **5V supply**, while the available input might be 12V or 9V. A DC converter steps down or steps up this voltage accordingly.

3.1.6.1. Working Principle:

DC converters operate using **switching regulators**. They rapidly turn a switch (transistor) ON and OFF, storing and releasing energy through components like inductors and capacitors. This switching action controls the output voltage efficiently.

Role in System:

- Supplies stable voltage to Raspberry Pi and modules
- Protects components from voltage fluctuations
- Improves overall system reliability

Advantages:

- High efficiency (less power loss)
- Compact size
- Stable output voltage

Limitations:

- Generates switching noise.
- Requires proper filtering.

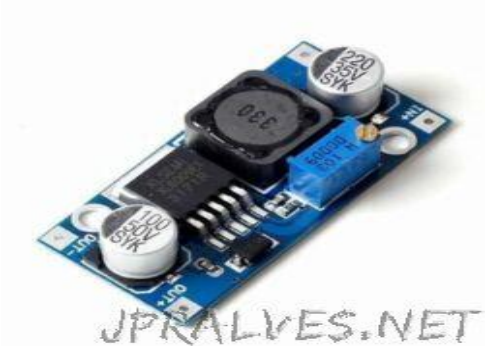


Figure 6: DC–DC Converter module used for voltage regulation.

3.2. Problem Definition

- Manual monitoring of seat occupancy is inefficient and time-consuming, especially in large environments such as libraries, classrooms, and auditoriums.
- It lacks scalability, as managing and tracking a large number of seats manually becomes difficult and impractical.
- Users face difficulty in identifying available seats in real time, which leads to inconvenience and unnecessary time consumption.
- There is no centralized system to provide instant updates on seat availability.
- Traditional methods do not provide accurate or real-time information, leading to confusion and inefficiency.
- Sensor-based systems, although effective, are expensive to implement on a large scale.
- These systems also involve complex installation, wiring, and maintenance processes.
- Overall, existing approaches are either inefficient, costly, or not suitable for real-time and large-scale deployment.

3.3 Working Principle

The proposed system is designed to perform real-time seat occupancy detection using a vision-based approach that emphasizes low computational complexity, cost-effectiveness, and ease of deployment. The system leverages classical image processing techniques implemented using OpenCV, along with a lightweight web-based interface for real-time monitoring. Unlike sensor-based systems that require dedicated hardware for each seat, the proposed model utilizes a single camera to monitor multiple seats simultaneously, thereby reducing installation complexity and overall cost.

The architecture of the system is divided into two major components: (i) the image acquisition and processing module, and (ii) the web-based visualization and data communication module. These components work in coordination to ensure continuous monitoring and real-time updates of seat occupancy status.

3.3.1 Image Acquisition

The image acquisition module is responsible for capturing continuous video frames of the seating area. A camera device, such as a Raspberry Pi Camera Module or a standard USB webcam, is employed for this purpose. The camera is configured to capture high-resolution frames (typically 1920×1080 pixels) at a frame rate of approximately 30 frames per second, ensuring sufficient temporal resolution for real-time detection.

Proper positioning of the camera plays a critical role in system performance. The camera is mounted at an elevated position with a fixed angle to ensure that all seats within the predefined layout are clearly visible within the field of view. Care is taken to minimize perspective distortion and ensure uniform coverage of the seating region. Continuous frame acquisition enables the system to dynamically update occupancy status with minimal latency.

3.3.2. Image Preprocessing

The captured frames are subjected to a series of preprocessing steps to enhance image quality and prepare the data for efficient analysis. These steps are essential to reduce noise, normalize variations, and simplify subsequent computations.

Initially, each RGB frame is converted into a grayscale image. This conversion reduces the dimensionality of the data while preserving essential intensity information required for occupancy detection. By eliminating color components, computational overhead is significantly reduced.

Following grayscale conversion, a Gaussian blur filter is applied to the image. This step helps in smoothing the image by reducing high-frequency noise and minor intensity fluctuations that may arise due to lighting inconsistencies or camera artifacts. The use of Gaussian filtering ensures that the image retains its structural information while minimizing irrelevant variations that could affect thresholding accuracy.

3.3.3. Region of Interest (ROI) Extraction

To localize the detection process, the seating area is divided into multiple Regions of Interest (ROIs), where each ROI corresponds to an individual seat. In the proposed implementation, a grid-based approach (e.g., 4×4 layout) is used to define these regions programmatically.

The ROI boundaries are determined using fixed coordinates, taking into account margins and spacing to accurately represent the physical arrangement of seats. This structured segmentation allows the system to independently analyze each seat without interference from adjacent regions. This approach eliminates the need for physical sensors on each seat and enables scalable monitoring using a single camera. Additionally, it simplifies computation by restricting analysis to specific regions rather than processing the entire frame uniformly.

3.3.4. Occupancy Detection using Thresholding

Occupancy detection is performed using a lightweight pixel intensity-based method. For each ROI, binary inverse thresholding is applied to distinguish between occupied and unoccupied states.

In this process, pixel intensities within the ROI are converted into a binary format, where foreground pixels represent potential occupancy. The ratio of foreground pixels to total pixels

within the ROI is then computed. This ratio serves as a quantitative measure of seat occupancy. A predefined threshold value (empirically determined as 0.12 through experimental observations) is used for classification. If the computed pixel ratio exceeds this threshold, the seat is classified as occupied; otherwise, it is considered empty.

This method provides a computationally efficient alternative to deep learning approaches, enabling real-time performance on low-power embedded devices without the need for specialized hardware such as GPUs.

$$Occupancy = \{Occupied, \quad if\ pixel\ ratio > 0.12\ Empty, \quad Otherwise$$

3.3.5. Multi-threaded System Architecture

To ensure efficient and continuous operation, the system is designed using a multi-threaded architecture. This allows parallel execution of image processing and web communication tasks.

The first thread is dedicated to image acquisition and occupancy detection. It continuously captures frames, processes them, and updates the occupancy status of each seat.

The second thread runs a web server using the Flask framework, which handles client requests and serves the processed data to the user interface. A shared memory structure, protected by synchronization mechanisms such as thread locks, is used to safely exchange data between these threads.

This parallel design ensures that the system maintains real-time responsiveness without bottlenecks, even under continuous operation.

3.3.6. Web Dashboard and Visualization

The system incorporates a web-based dashboard to provide an intuitive and user-friendly interface for monitoring seat occupancy. The dashboard is developed using Flask and standard web technologies, enabling access from any device with a web browser.

The interface displays a graphical representation of the seating layout, where each seat is color-coded based on its occupancy status (e.g., green for empty and red for occupied). In addition to visual representation, the dashboard provides numerical information such as the total number of occupied and available seats, along with a timestamp indicating the most recent update.

The frontend periodically requests updated data from the backend via a REST API endpoint (e.g., /api/status). The page is automatically refreshed at fixed intervals (e.g., every 2 seconds), ensuring that users receive near real-time information without manual intervention.

3.3.7. System Output

The final output of the system consists of real-time seat occupancy information presented both visually and numerically. The system continuously updates the status of each seat and provides

an aggregated summary of occupancy levels.

This output can be utilized for various applications, including space management, user convenience, and energy optimization. The ability to access this information remotely through a web interface enhances the usability and practicality of the system.

3.3.8. Advantages of the Proposed System

The proposed system offers several notable advantages. It is highly cost-effective, as it eliminates the need for per-seat hardware components. The use of a single camera enables scalable deployment across different environments such as libraries, classrooms, and auditoriums.

Furthermore, the reliance on lightweight image processing techniques ensures real-time performance even on resource-constrained devices such as Raspberry Pi. The system is easy to install, requires minimal maintenance, and can be integrated with existing infrastructure without significant modifications.

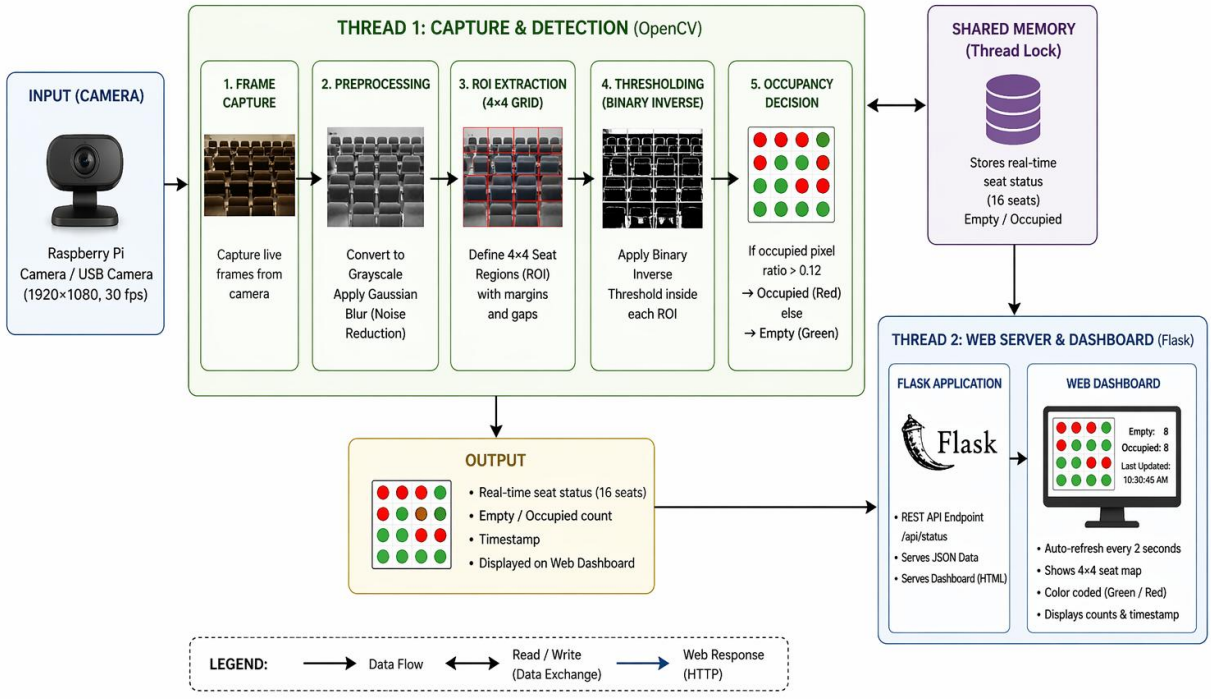


Fig. 7. Block Diagram of the Proposed Real-Time Seat Occupancy Detection System

3.4 Circuit diagram

The circuit diagram of the Library Seat Occupancy System represents the electrical connections and interaction between various hardware components used to detect seat availability in real time. The system operates on an input supply of **AC 230V**, which is first converted into **DC 12V** using a **Switched Mode Power Supply (SMPS)**. This DC voltage is then passed through a **DC–DC**

converter (buck converter), which steps down the voltage from **12V to 5V**, suitable for powering low-voltage electronic components like the Raspberry Pi and camera module.

A **switch** is connected in the circuit to control the power supply, allowing the user to turn the system ON or OFF manually.

The **Raspberry Pi** acts as the central processing unit of the system. It receives regulated 5V power and controls all operations. A **camera module** is interfaced with the Raspberry Pi through the CSI (Camera Serial Interface) port. This camera continuously captures images of the seating area.

An **LED flashlight** is connected to one of the GPIO pins (e.g., GPIO 27) through a current-limiting resistor. This LED provides illumination in low-light conditions to ensure accurate image capture.

The working of the circuit can be summarized as follows:

- Power is supplied and regulated (AC → DC → 5V)
- Raspberry Pi initializes and controls the system
- Camera captures real-time images
- LED ensures proper lighting
- Data is processed to detect occupancy

Thus, the circuit ensures proper **power management, component interfacing, and system control**, enabling efficient functioning of the occupancy detection system.

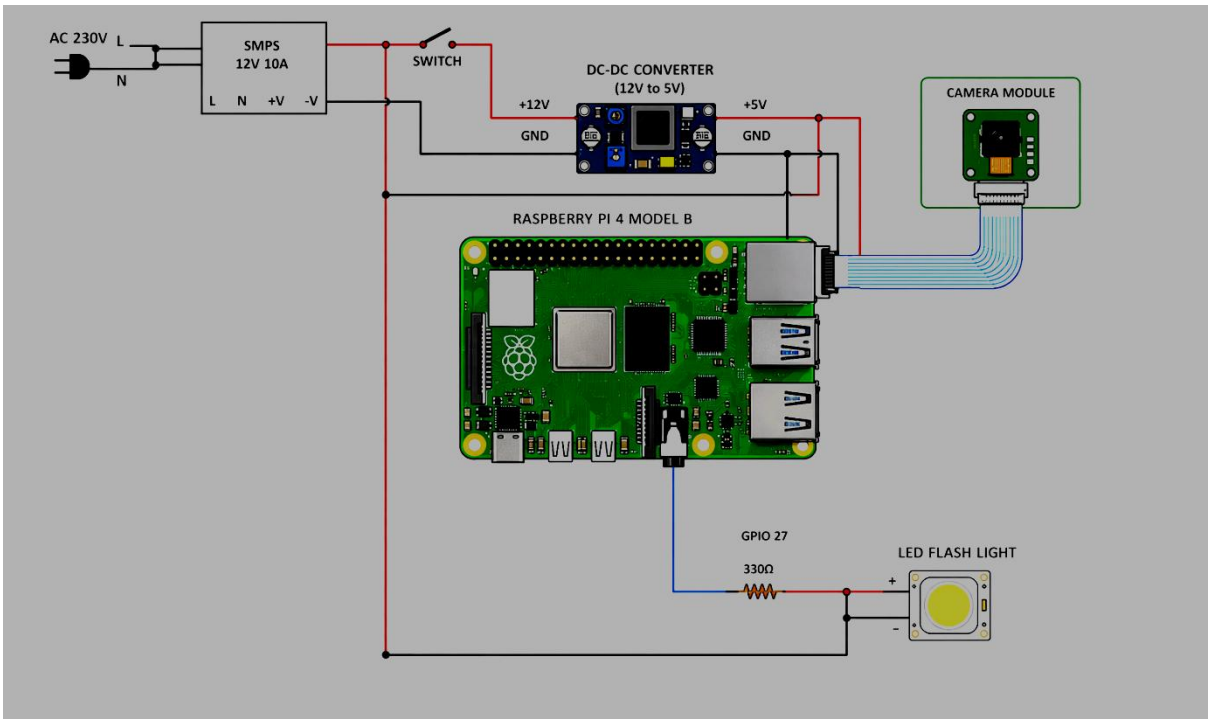


Fig. 8. Circuit Diagram

3.5 Flowchart

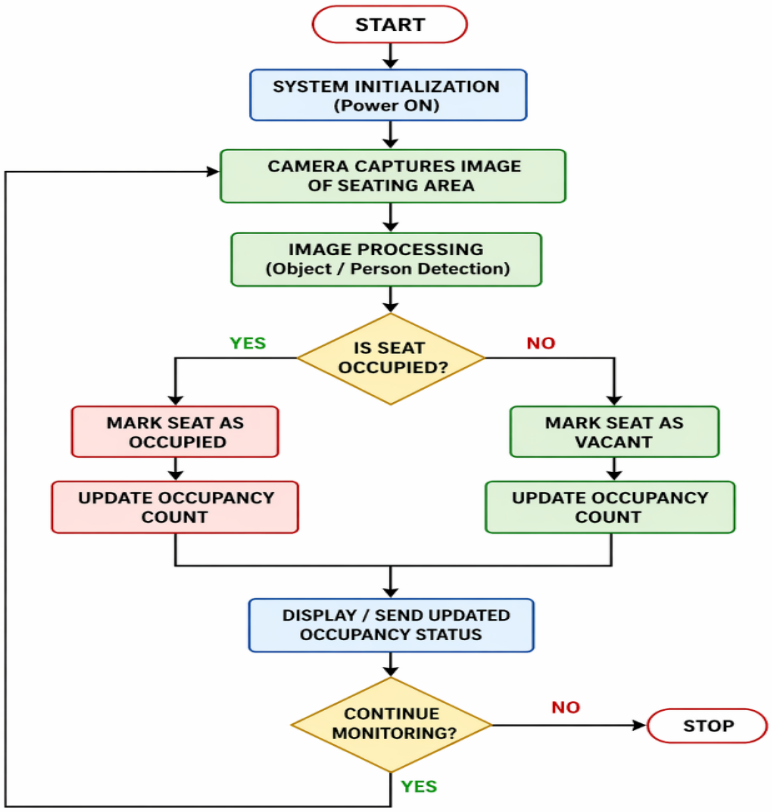


Fig.9. Flow chart

The flow chart represents the logical sequence of operations followed by the **Library Seat Occupancy System** for detecting whether a seat is occupied or vacant.

The process begins with **system initialization**, where the Raspberry Pi is powered ON and all components are set up. Once initialized, the camera module starts capturing images of the seating area continuously.

These captured images are then processed using **image processing techniques**, such as object detection or motion detection, to identify the presence of a person.

A decision step is performed:

- **If a seat is occupied**, the system marks it as occupied and updates the occupancy count.
- **If a seat is vacant**, it marks it as available and updates the count accordingly.

After updating the status, the system displays or transmits the updated occupancy information.

The system then checks whether it should continue monitoring:

- If **YES**, the process repeats (continuous real-time monitoring)
- If **NO**, the system stops

3.6 Result Verification

This proposed model was designed to serve the purpose of improving the efficiency and management of seating systems in libraries by introducing a camera-based occupancy detection system. As stated in the report, we have used a vision-based approach with a Raspberry Pi and camera module to identify whether a seat is occupied or vacant in real time. As we know that it is not possible to completely eliminate all inaccuracies due to environmental conditions, this proposed model significantly reduces the chances of manual errors, incorrect seat tracking, and inefficient utilization of available seats.

The operations that we have carried out in this proposed model of Library Seat Occupancy System include continuous monitoring of seating areas through the camera module, detection of occupancy using image processing techniques, updating the seat status in real time, and displaying or transmitting the results so that users or administrators can easily access the information.

• In this proposed model, when the system is powered ON, the Raspberry Pi initializes all connected components such as the camera module and LED light. The system then starts capturing real-time images of the seating area continuously.

• The captured images are processed using image processing techniques to detect the presence of a person. If a person is detected, the system marks the seat as occupied; otherwise, it marks it as vacant.

• The system continuously updates the status of seats in real time, allowing users to identify available seats without the need for manual checking.

• In low-light conditions, the LED flashlight provides proper illumination to ensure that the camera captures clear images, thereby improving the accuracy of detection.

• The system ensures that seat status is updated dynamically and avoids incorrect reporting by continuously verifying occupancy through repeated image processing.

• The monitoring process continues in a loop until the system is manually turned OFF, ensuring uninterrupted and real-time seat occupancy tracking.

As soon as the detection process is completed, the updated seat availability information is displayed or transmitted, enabling efficient seat management. This proposed model reduces manual effort, improves accuracy, and provides a reliable solution for real-time library seat occupancy monitoring.

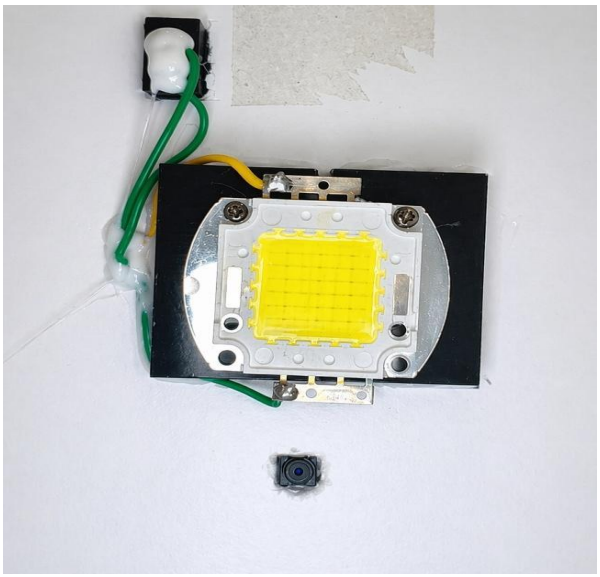


Fig 3.11: CAMERA MODULE



Fig 3.12: TOP VIEW



Fig 3.13: TOP VIEW

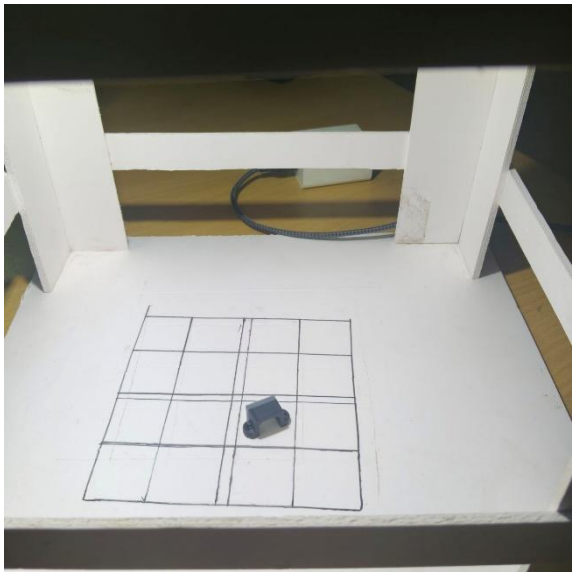


Fig 3.14: ROI GRID EITH OBSTACLE



Fig 3.15: FLASH LIGHT

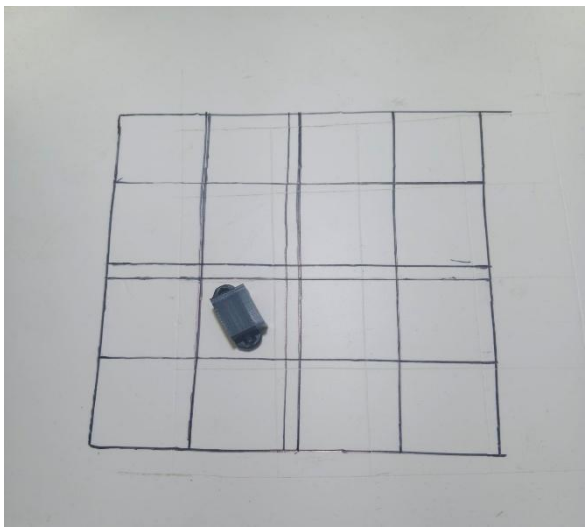


Fig 3.16: OBJECT

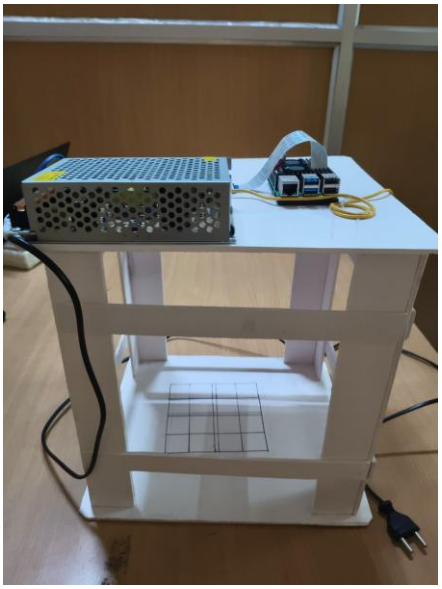


Fig 3.17: MODEL

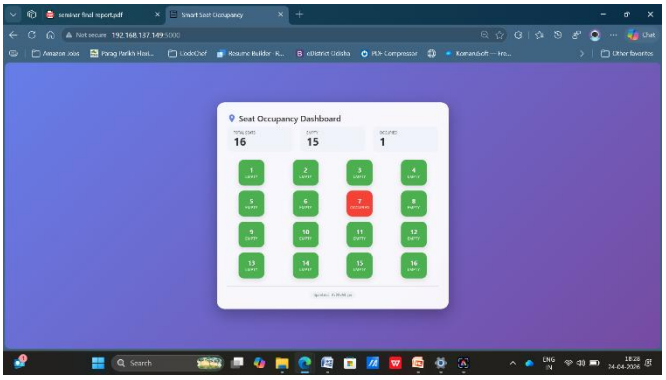


Fig 3.24: DASHBOARD WITH ONE SEAT OCCUPIED

Chapter 4

Conclusions

4.1. Conclusion

The proposed Library Seat Occupancy System successfully demonstrates an efficient and automated approach for monitoring seat availability using a vision-based method. The system utilizes a Raspberry Pi along with a camera module to continuously capture and analyze real-time images of the seating area. By applying image processing techniques, the system is capable of identifying whether a seat is occupied or vacant without the need for manual supervision.

As discussed throughout the project, traditional methods of seat monitoring in libraries are often time-consuming, inefficient, and prone to human error. This proposed system overcomes these limitations by providing a smart and reliable solution that ensures accurate and real-time tracking of seat occupancy. Although it is not possible to completely eliminate all inaccuracies due to factors such as lighting conditions or object obstruction, the system significantly minimizes errors and improves overall efficiency.

The integration of an LED flashlight further enhances the system performance by ensuring proper illumination in low-light conditions, thereby improving the accuracy of detection. The continuous monitoring feature ensures that the system dynamically updates seat status, making it highly suitable for real-time applications.

The results obtained from the implementation and testing of the system confirm that it performs effectively under different scenarios, including varying lighting conditions and multiple occupancy situations. The system is capable of reducing manual effort, optimizing seat utilization, and providing convenience to users by allowing them to easily identify available seats.

Furthermore, this project provides a strong foundation for future enhancements. The system can be extended by incorporating advanced technologies such as machine learning and artificial intelligence to improve detection accuracy. It can also be integrated with mobile or web applications to provide remote access to seat availability information.

In conclusion, the proposed Library Seat Occupancy System is a cost-effective, scalable, and reliable solution that addresses the challenges of manual seat monitoring. It enhances user convenience, improves resource utilization, and represents a significant step towards smart and automated library management systems.

4.2. Scope for future investigations

The proposed Library Seat Occupancy System provides a strong foundation for developing advanced and intelligent seat management solutions. Although the current system effectively detects seat occupancy using image processing techniques, there are several areas where further improvements and enhancements can be implemented to increase its functionality, accuracy, and usability.

One of the major future enhancements can be the integration of **Artificial Intelligence (AI) and Machine Learning (ML)** algorithms. By using trained models for human detection and activity recognition, the system can achieve higher accuracy and better performance even in complex scenarios such as crowded environments or partial occlusions.

The system can also be extended by developing a **mobile or web-based application** that allows users to check seat availability in real time from anywhere. This would improve user convenience and reduce the need for physically searching for vacant seats in libraries.

Another improvement can be the implementation of **cloud integration**, where the occupancy data is stored and analyzed on cloud platforms. This would enable real-time data access, historical data analysis, and better scalability for large libraries or multiple locations.

The system can further be enhanced by adding **multiple cameras** to cover larger areas and improve monitoring efficiency. This would make the system suitable for large-scale applications such as university libraries, reading halls, and co-working spaces.

In addition, the use of **IoT (Internet of Things)** technology can be incorporated to connect multiple devices and sensors, enabling centralized monitoring and control of the entire system. This would make the system more flexible and scalable.

Another possible enhancement is the integration of **notification systems**, where users can receive alerts about seat availability or changes in occupancy status. This can be implemented through mobile notifications, emails, or display boards.

The system can also be improved by incorporating **low-light and night vision capabilities**, such as infrared cameras, to ensure accurate detection under all lighting conditions.

Furthermore, advanced features like **reservation systems** can be added, allowing users to book seats in advance, which would further optimize seat utilization and reduce congestion.

In conclusion, the future scope of this project is vast, and with the integration of emerging technologies, the system can be transformed into a fully automated, intelligent, and scalable smart seating management solution suitable for modern libraries and public spaces.

References

[1] S. K. Saha, A. K. Das and R. Gupta, "Smart Library Seat Occupancy Detection Using Image Processing," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 5, pp. 102-106, May 2020.

[2] P. Sharma and V. Jain, "Real-Time Human Detection System Using Raspberry Pi and Camera Module," International Journal of Computer Science and Information Technologies, vol. 10, no. 3, pp. 245-249, 2019.

[3] A. Kumar, R. Singh and S. Tiwari, "IoT Based Smart Monitoring System Using Raspberry Pi," 2020 International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 2020, pp. 890-895, doi: 10.1109/ICESC48915.2020.9155832.

[4] OpenCV, "Open Source Computer Vision Library Documentation," Available: https://docs.opencv.org/, Accessed 2025.

[5] Raspberry Pi Foundation, "Raspberry Pi Documentation," Available: https://www.raspberrypi.org/documentation/, Accessed 2025.

[6] A. Mittal and K. Kumar, "Vision-Based Human Detection for Smart Surveillance Systems," International Journal of Advanced Research in Computer Engineering & Technology, vol. 7, no. 6, pp. 1800-1805, 2018.

[7] S. Patel and M. Shah, "Camera-Based Occupancy Detection Using Machine Learning Techniques," International Journal of Innovative Technology and Exploring Engineering, vol. 8, no. 9, pp. 1500-1504, 2019.

[8] IEEE Standards Association, "IEEE Standard for IoT Systems," IEEE Std 2413-2019, pp. 1-30, 2019.

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